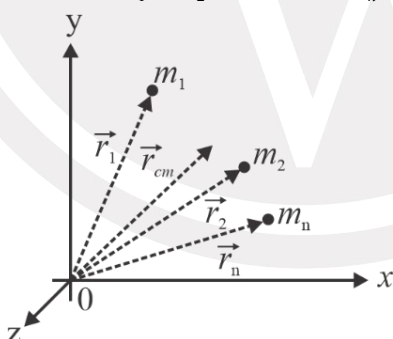
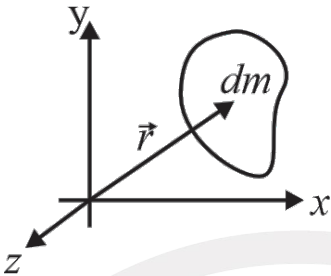
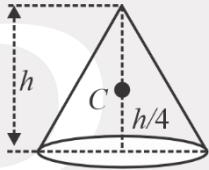
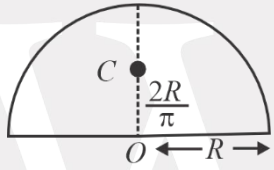
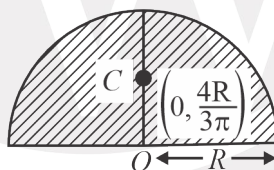
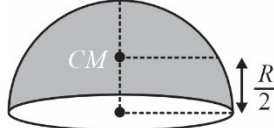
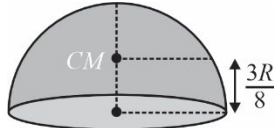
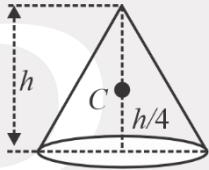
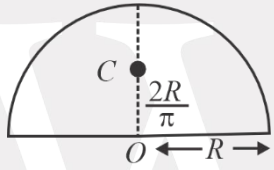
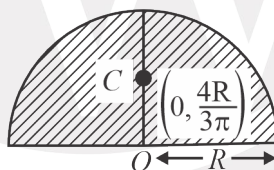
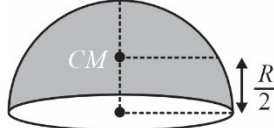
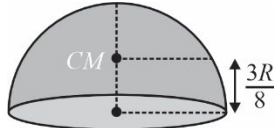
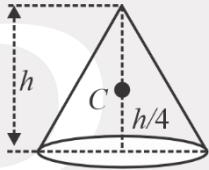
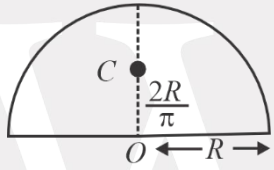
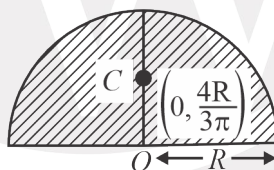
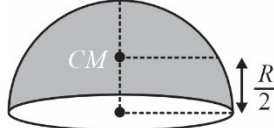
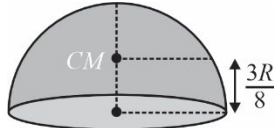
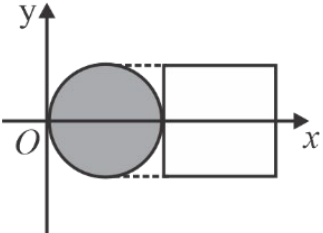
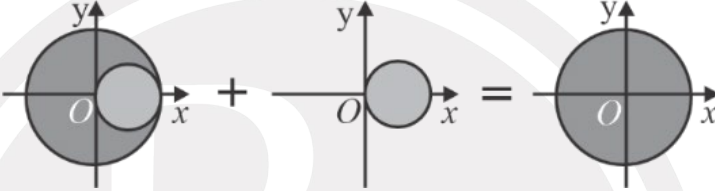
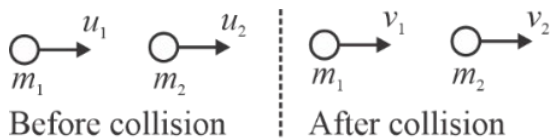
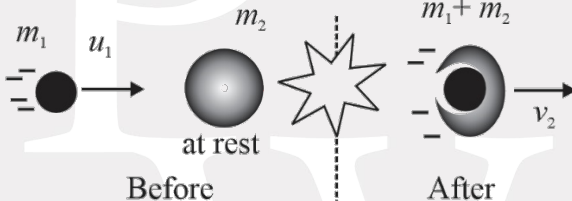


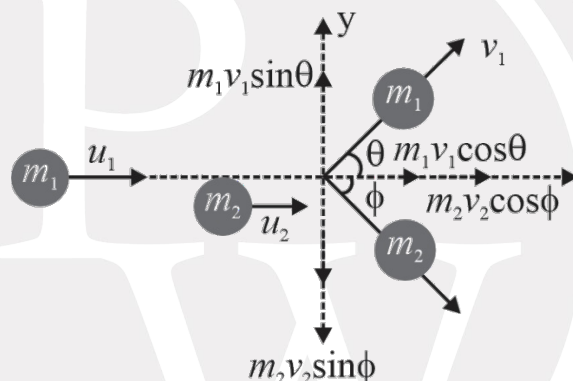
# Centre of Mass and System of Particles

| Sr. No. | Concept                                | Formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Other information                                                                                                                                                                                                           |
|---------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.      | Centre of mass of discrete mass system | <p>Position vector of center of mass of all <math>n</math>-particles</p> $\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$ $\vec{r}_{cm} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + \dots + m_n \vec{r}_n}{m_1 + m_2 + m_3 + \dots + m_n}$ $\vec{r}_{cm} = x_{cm} \hat{i} + y_{cm} \hat{j} + z_{cm} \hat{k}$ <p>Where</p> $x_{cm} = \frac{m_1 x_1 + m_2 x_2 + \dots + m_n x_n}{m_1 + m_2 + \dots + m_n}$ $y_{cm} = \frac{m_1 y_1 + m_2 y_2 + \dots + m_n y_n}{m_1 + m_2 + \dots + m_n}$  $z_{cm} = \frac{m_1 z_1 + m_2 z_2 + \dots + m_n z_n}{m_1 + m_2 + \dots + m_n}$ <p>For a system of two particles</p> $r \propto \frac{1}{m} \Rightarrow m_1 r_1 = m_2 r_2$ <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="border: 1px solid black; padding: 5px;"> <math>r_1 = \frac{m_2 r_2}{m_1}</math> </div> <div>and</div> <div style="border: 1px solid black; padding: 5px;"> <math>r_2 = \frac{m_1 r_1}{m_2}</math> </div> </div> <p><math>r_1</math> and <math>r_2</math> are distances of <math>m_1</math> and <math>m_2</math> from center of mass</p> | <p><math>r_{cm}</math> = position of centre of mass from origin.</p> <p><math>\vec{r}_1, \vec{r}_2, \vec{r}_3, \dots, \vec{r}_n</math> = position vectors of masses <math>m_1, m_2, m_3, \dots, m_n</math> respectively</p> |

| Sr. No.                                  | Concept                                                                             | Formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Other information                                                                                         |        |                |                 |                                                                                   |                                                                                                    |                           |                                                                                     |                                                      |                                    |                                                                                     |                                                       |                                          |                                                                                     |                                                   |                  |                                                                                     |                                                    |                                                                           |
|------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------|----------------|-----------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------|------------------|-------------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------|
| 2.                                       | Centre of mass of continuous mass distribution                                      | $x_{cm} = \frac{\int x dm}{\int dm}, \quad y_{cm} = \frac{\int y dm}{\int dm}, \quad z_{cm} = \frac{\int z dm}{\int dm}$ <p>The algebraic sum of the moments of masses of various particles of a system about its centre of mass is zero.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $dm$ = small mass element of a continuous body.<br>$\vec{r}$ = position vector of small mass from origin. |        |                |                 |                                                                                   |                                                                                                    |                           |                                                                                     |                                                      |                                    |                                                                                     |                                                       |                                          |                                                                                     |                                                   |                  |                                                                                     |                                                    |                                                                           |
| 3.                                       | Position of centre of mass of some special bodies                                   | <table><tr><th>Body</th><th>Figure</th><th>Position of CM</th></tr><tr><td>Cone or pyramid</td><td></td><td>On the axis of the cone at a distance <math>3h/4</math> from the vertex, where <math>h</math> is the height of the cone.</td></tr><tr><td>Uniform semicircular ring</td><td></td><td>Coordinates of <math>CM = \left(0, \frac{2R}{\pi}\right)</math></td></tr><tr><td>Uniform semicircular plate or disc</td><td></td><td>Coordinates of <math>CM = \left(0, \frac{4R}{3\pi}\right)</math></td></tr><tr><td>Hemispherical shell or hollow hemisphere</td><td></td><td>Coordinates of <math>CM = \left(0, \frac{R}{2}\right)</math></td></tr><tr><td>Solid hemisphere</td><td></td><td>Coordinates of <math>CM = \left(0, \frac{3R}{8}\right)</math></td></tr></table> | Body                                                                                                      | Figure | Position of CM | Cone or pyramid |  | On the axis of the cone at a distance $3h/4$ from the vertex, where $h$ is the height of the cone. | Uniform semicircular ring |  | Coordinates of $CM = \left(0, \frac{2R}{\pi}\right)$ | Uniform semicircular plate or disc |  | Coordinates of $CM = \left(0, \frac{4R}{3\pi}\right)$ | Hemispherical shell or hollow hemisphere |  | Coordinates of $CM = \left(0, \frac{R}{2}\right)$ | Solid hemisphere |  | Coordinates of $CM = \left(0, \frac{3R}{8}\right)$ | $C$ = centre of mass position here.<br>$R$ = radius of semicircular ring. |
| Body                                     | Figure                                                                              | Position of CM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                           |        |                |                 |                                                                                   |                                                                                                    |                           |                                                                                     |                                                      |                                    |                                                                                     |                                                       |                                          |                                                                                     |                                                   |                  |                                                                                     |                                                    |                                                                           |
| Cone or pyramid                          |    | On the axis of the cone at a distance $3h/4$ from the vertex, where $h$ is the height of the cone.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |        |                |                 |                                                                                   |                                                                                                    |                           |                                                                                     |                                                      |                                    |                                                                                     |                                                       |                                          |                                                                                     |                                                   |                  |                                                                                     |                                                    |                                                                           |
| Uniform semicircular ring                |  | Coordinates of $CM = \left(0, \frac{2R}{\pi}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                           |        |                |                 |                                                                                   |                                                                                                    |                           |                                                                                     |                                                      |                                    |                                                                                     |                                                       |                                          |                                                                                     |                                                   |                  |                                                                                     |                                                    |                                                                           |
| Uniform semicircular plate or disc       |  | Coordinates of $CM = \left(0, \frac{4R}{3\pi}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                           |        |                |                 |                                                                                   |                                                                                                    |                           |                                                                                     |                                                      |                                    |                                                                                     |                                                       |                                          |                                                                                     |                                                   |                  |                                                                                     |                                                    |                                                                           |
| Hemispherical shell or hollow hemisphere |  | Coordinates of $CM = \left(0, \frac{R}{2}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                           |        |                |                 |                                                                                   |                                                                                                    |                           |                                                                                     |                                                      |                                    |                                                                                     |                                                       |                                          |                                                                                     |                                                   |                  |                                                                                     |                                                    |                                                                           |
| Solid hemisphere                         |  | Coordinates of $CM = \left(0, \frac{3R}{8}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |        |                |                 |                                                                                   |                                                                                                    |                           |                                                                                     |                                                      |                                    |                                                                                     |                                                       |                                          |                                                                                     |                                                   |                  |                                                                                     |                                                    |                                                                           |

| Sr. No. | Concept                            | Formulas                                                                                                                                                                                                                                                                                                                                                      | Other information                                                                                                                                                                                                                                                                          |
|---------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.      | Centre of Mass of composite bodies | <p>Centre of mass of the disc and square composite body:</p>  <p>Co-ordinates of centre of mass of composite body are</p> $x_c = \frac{m_d x_d + m_s x_s}{m_d + m_s} \text{ and } y_c = \frac{m_d y_d + m_s y_s}{m_d + m_s}$                                                 | <p><math>m_d</math> = mass of disc<br/> <math>m_s</math> = mass of square.<br/> <math>x_d</math> and <math>x_s</math> = x - coordinates of disc and square respectively<br/> <math>y_d</math> and <math>y_s</math> = y - coordinates of centre of mass of disc and square respectively</p> |
| 5.      | Centre of mass of truncated bodies |  <p>The formula changes to:</p> $x_{cm} = \frac{Mx - mx'}{M - m}; y_{cm} = \frac{My - my'}{M - m}; z_{cm} = \frac{Mz - mz'}{M - m}$ <p>Where, <math>x'</math>, <math>y'</math> and <math>z'</math> represent the coordinates of the centre of mass of the removed part.</p> | <p><math>M</math> = mass of the complete body<br/> <math>m</math> = mass of the removed part of the body.</p>                                                                                                                                                                              |
| 6.      | Coefficient of restitution ( $e$ ) | $e = \frac{v_2 - v_1}{u_1 - u_2}$ <p>For a perfectly elastic collision, <math>e = 1</math><br/> For a perfectly inelastic collision, <math>e = 0</math><br/> For all other collisions, <math>0 &lt; e &lt; 1</math></p>                                                                                                                                       | <p><math>v_2 - v_1</math> = relative velocity of separation<br/> <math>u_1 - u_2</math> = relative velocity of approach.</p>                                                                                                                                                               |

| Sr. No. | Concept                                                                  | Formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Other information                                                                                                                                                    |
|---------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.      | Elastic collision                                                        |  <p>Before collision      After collision</p> <p>(i) Linear momentum is conserved<br/> <math display="block">m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2</math></p> <p>(ii) Kinetic energy is conserved in elastic collisions.<br/> <math display="block">\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2</math></p> <p>(iii) In one-dimensional elastic collision, velocities after the collision are given by<br/> <math display="block">v_1 = \frac{m_1 - m_2}{m_1 + m_2} \cdot u_1 + \frac{2m_2}{m_1 + m_2} \cdot u_2</math> <math display="block">v_2 = \frac{2m_1}{m_1 + m_2} \cdot u_1 + \frac{m_2 - m_1}{m_1 + m_2} \cdot u_2</math></p>                                                  | $m_1$ and $m_2$ = mass of the two bodies.<br>$u_1$ and $u_2$ = initial velocities of the two bodies.<br>$v_1$ and $v_2$ = final velocities of the two bodies.        |
| 8.      | Head on perfectly inelastic collision (two bodies stick after collision) |  <p>Before      After</p> <p>Momentum      <math>m_1 u_1</math>      <math>(m_1 + m_2) v_2</math></p> <p>Kinetic energy      <math>\frac{1}{2} m_1 u_1^2</math>      <math>\frac{1}{2} (m_1 + m_2) v_2^2</math></p> <p>From conservation of momentum:<br/> <math display="block">m_1 u_1 = (m_1 + m_2) v_2 \Rightarrow v_2 = \frac{m_1}{m_1 + m_2} u_1</math></p> <p>Ratio of kinetic energies before and after collision:<br/> <math display="block">\frac{KE_f}{KE_i} = \frac{m_1}{m_1 + m_2}</math></p> <p>Fraction of kinetic energy lost in the collision:<br/> <math display="block">\frac{KE_i - KE_f}{KE_i} = \frac{m_2}{m_1 + m_2}</math></p> <p>Loss of <math>KE = KE_i - KE_f = \frac{m_1 m_2 u_1^2}{2(m_1 + m_2)}</math></p> | $KE_f$ = final kinetic energy.<br>$KE_i$ = initial kinetic energy.<br>$v_2$ = final velocity after Collision.<br>$u_1$ = initial velocity of $m_1$ before collision. |

| Sr. No. | Concept                               | Formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Other information                                                                                                                                                                                                                                                                                                                        |
|---------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9.      | Bouncing of ball                      | <p>Velocity immediately after the <math>n^{\text{th}}</math> rebound:</p> $v_n = e^n \sqrt{2gh}$ <p>Height attained by the ball after the <math>n^{\text{th}}</math> rebound:</p> $h_n = e^{2n} h$ <p>Total time elapsed before the ball stop:</p> $T = \sqrt{\frac{2h}{g}} \left( \frac{1+e}{1-e} \right)$ <p>Distance covered by the ball before it stops:</p> $s = h \left( \frac{1+e^2}{1-e^2} \right)$                                                    | <p><math>v_1, v_2, v_3, \dots, v_n</math> = the velocities immediately after first, second, third, ..., <math>n^{\text{th}}</math> collisions with the ground.</p> <p><math>h</math> = height from which the ball is dropped.</p> <p><math>e</math> = coefficient of restitution</p> <p><math>g</math> = gravitational acceleration.</p> |
| 10.     | Oblique collision (elastic collision) |  <p>By conservation of momentum</p> <p><math>x</math>-axis: <math>m_1 u_1 + m_2 u_2 = m_1 v_1 \cos \theta + m_2 v_2 \cos \phi</math></p> <p><math>y</math>-axis: <math>0 + 0 = m_1 v_1 \sin \theta - m_2 v_2 \sin \phi</math></p> <p>By conservation of kinetic energy:</p> $\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$ | <p><math>m_1</math> and <math>m_2</math> = mass of the two particles.</p> <p><math>u_1</math> and <math>u_2</math> = initial velocity of particles.</p> <p><math>v_1</math> and <math>v_2</math> = final velocity of particles after collision.</p>                                                                                      |